

Formulari. Electrònica Física

Equació de Schrödinger:

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \Psi = E \Psi$$

$$\Psi_n(x) = A_n \sin\left(\frac{n\pi x}{a}\right)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Densitat d'estats:

$$g_c(E) = \frac{m_n^* \sqrt{2m_n^*(E - E_c)}}{\pi^2 \hbar^3} \quad E \geq E_c$$
$$g_v(E) = \frac{m_p^* \sqrt{2m_p^*(E_v - E)}}{\pi^2 \hbar^3} \quad E \leq E_v$$

Probabilitat de què un estat disponible d'energia E sigui ocupat:

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

$$n = N_c e^{-(E_c - E_F)/kT} \quad \text{on} \quad N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$$

$$p = N_v e^{-(E_F - E_v)/kT} \quad \text{on} \quad N_v = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2}$$

A un semiconductor intrínsec ($n = p = n_i$):

$$n = N_c e^{(E_i - E_c)/kT} = N_v e^{(E_v - E_i)/kT} = p$$

$$n_i^2 = N_c N_v \exp(-E_g/kT)$$

Al silici:

$$E_g = 1.17 \text{ eV} - 4.73 \times 10^{-4} \frac{T^2}{T + 636}$$

Probabilitat d'ocupació d'un nivell donador

$$f(E_d) = \frac{1}{1 + \frac{1}{2} \exp[(E_d - E_F)/kT]} = \frac{N_d^0}{N_d}$$

Probabilitat d'ocupació d'un nivell acceptor:

$$f(E_a) = \frac{1}{1 + \frac{1}{2} \exp[(E_F - E_a)/kT]} = \frac{N_a^0}{N_a}$$

• $\mu_n \equiv [q\tau_{mn} / m_n^*]$ **mobilitat dels electrons**

• $\mu_p \equiv [q\tau_{mp} / m_p^*]$ **mobilitat dels forats**

$$J_N = J_{N,\text{drift}} + J_{N,\text{diff}} = qn\mu_n \mathcal{E} + qD_N \frac{dn}{dx}$$

$$J_P = J_{P,\text{drift}} + J_{P,\text{diff}} = qp\mu_p \mathcal{E} - qD_P \frac{dp}{dx}$$

$$D_N = \frac{kT}{q} \mu_n$$

$$D_P = \frac{kT}{q} \mu_p$$

$$\frac{d\Delta n}{dt} = \frac{d\Delta p}{dt} = - \frac{pn - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

on $n_1 \equiv n_i e^{(E_T - E_i)/kT}$ i $p_1 \equiv n_i e^{(E_i - E_T)/kT}$

**Equacions
de
continuitat:**

$$\begin{aligned} \frac{\partial n}{\partial t} &= \frac{1}{q} \frac{\partial J_n(x)}{\partial x} - \frac{\Delta n}{\tau_n} + G_L \\ \frac{\partial p}{\partial t} &= -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} - \frac{\Delta p}{\tau_p} + G_L \end{aligned}$$

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) + \frac{kT}{q} \ln\left(\frac{N_D}{n_i}\right) = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$I_0 = A q n_i^2 \left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right)$$

Amplada de depleció d'una unió pn (díode):

$$W = x_p + x_n = \sqrt{\frac{2\epsilon_s}{q} (V_{bi} - V_A) \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Díode amb base (zona n) curta

$$J_p = -q D_p \frac{d\Delta p_n}{dx} = q D_p \frac{n_i^2}{N_D} \left(e^{qV_A/kT} - 1 \right) \left(\frac{1}{x'_c} \right)$$

Díode amb base (zona n) llarga

$$J_p = q D_p \frac{n_i^2}{N_D} \left(e^{qV_A/kT} - 1 \right) \left(\frac{1}{L_p} \right)$$

BJT

Emissor i col·lector llargs:

$$I_E = qA \left[\left(\frac{D_E}{L_E} p_{E0} + \frac{D_B}{L_B} n_{B0} \frac{\cosh(W/L_B)}{\sinh(W/L_B)} \right) (e^{qV_{BE}/kT} - 1) - \left(\frac{D_B}{L_B} n_{B0} \frac{1}{\sinh(W/L_B)} \right) (e^{qV_{BC}/kT} - 1) \right]$$

$$I_C = qA \left[\left(\frac{D_B}{L_B} n_{B0} \frac{1}{\sinh(W/L_B)} \right) (e^{qV_{BE}/kT} - 1) - \left(\frac{D_C}{L_C} p_{C0} + \frac{D_B}{L_B} n_{B0} \frac{\cosh(W/L_B)}{\sinh(W/L_B)} \right) (e^{qV_{BC}/kT} - 1) \right]$$

Emissor i col·lector curts:

$$I_E = qA \left[\left(\frac{D_E}{L_E} p_{E0} + \frac{D_B}{L_B} n_{B0} \frac{\cosh(W/L_B)}{\sinh(W/L_B)} \right) (e^{qV_{BE}/kT} - 1) - \left(\frac{D_B}{L_B} n_{B0} \frac{1}{\sinh(W/L_B)} \right) (e^{qV_{BC}/kT} - 1) \right]$$

$$I_C = qA \left[\left(\frac{D_B}{L_B} n_{B0} \frac{1}{\sinh(W/L_B)} \right) (e^{qV_{BE}/kT} - 1) - \left(\frac{D_C}{L_C} p_{C0} + \frac{D_B}{L_B} n_{B0} \frac{\cosh(W/L_B)}{\sinh(W/L_B)} \right) (e^{qV_{BC}/kT} - 1) \right]$$

Model Ebers-Moll

$$I_C = \alpha_F I_{F0} (e^{qV_{EB}/kT} - 1) - I_{R0} (e^{qV_{CB}/kT} - 1)$$

$$I_E = I_{F0} (e^{qV_{EB}/kT} - 1) - \alpha_R I_{R0} (e^{qV_{CB}/kT} - 1)$$

$$\beta_{dc} = \frac{1}{\frac{n_{iE}^2}{n_{iB}^2} \frac{D_E}{D_B} \frac{N_B}{N_E} \frac{W}{L_E} + \frac{1}{2} \left(\frac{W}{L_B} \right)^2}$$

Amplada de depleció a un MOS amb substrat p

$$W_d = \sqrt{\frac{2\epsilon_{Si}\psi_s}{qN_A}}$$

Tensió llindar a un MOS amb substrat p

$$\psi_B = \frac{kT}{q} \ln(N_A / n_i)$$

$$V_T = V_{FB} + 2\psi_B + \frac{\sqrt{2qN_A\epsilon_{Si}(2\psi_B)}}{C_{ox}}$$